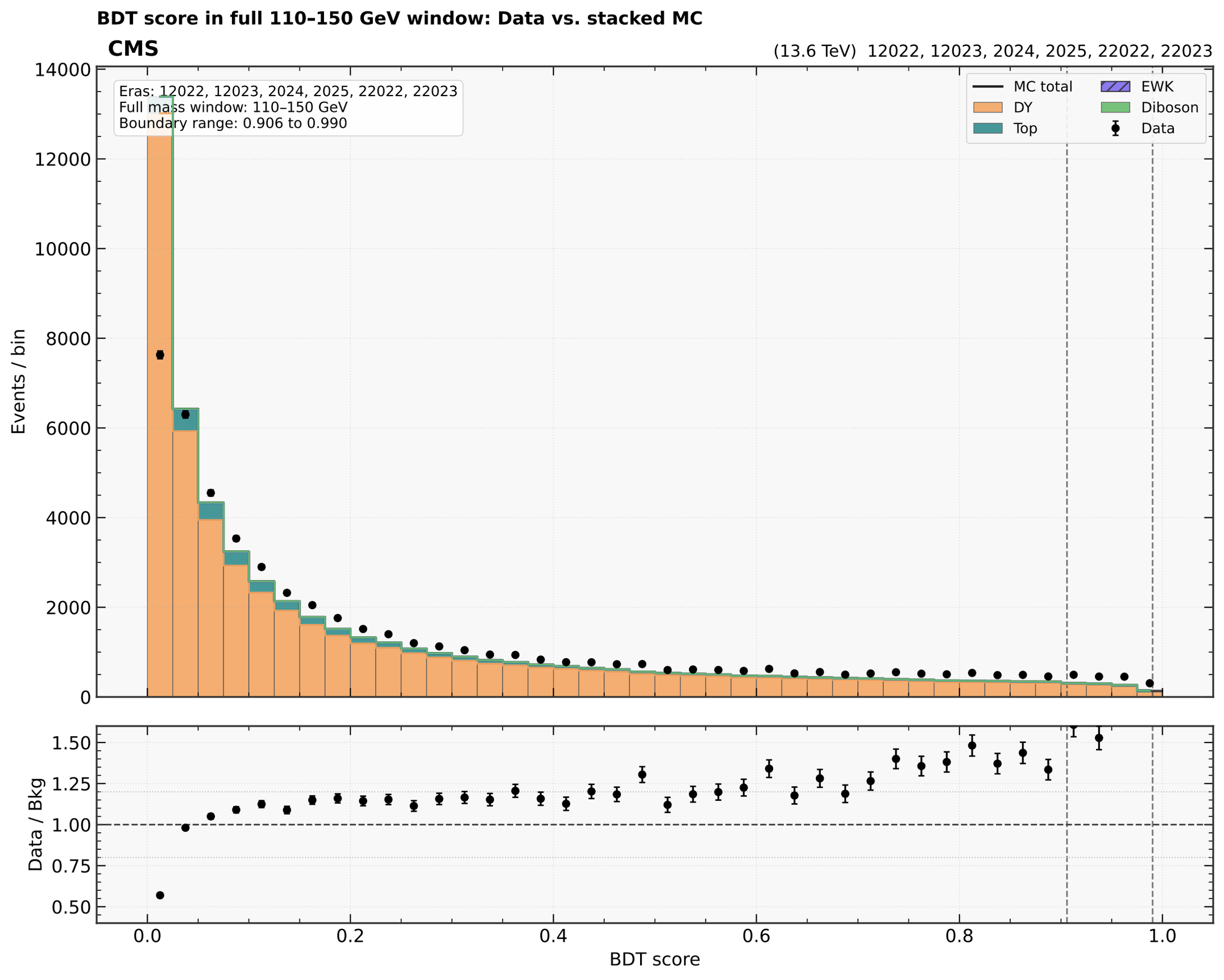
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022, 12023, 2024, 2025, 22022, 22023
Category edges : 0.0000, 0.9059, 0.9838, 0.9903, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

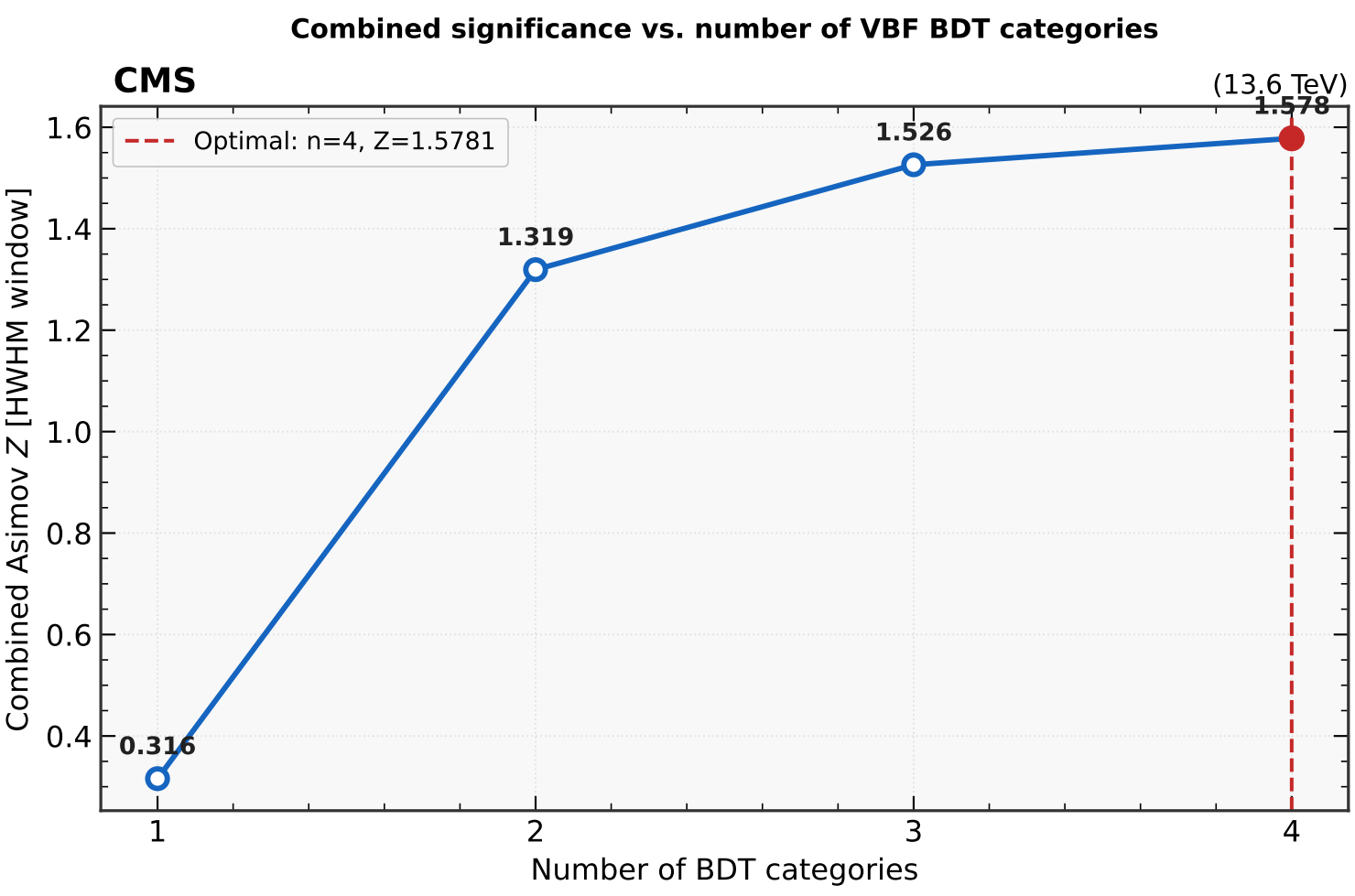

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
Eras: 12022, 12023, 2024, 2025, 22022, 22023
Full mass window: 110–150 GeV
Boundary range: 0.906 to 0.990
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

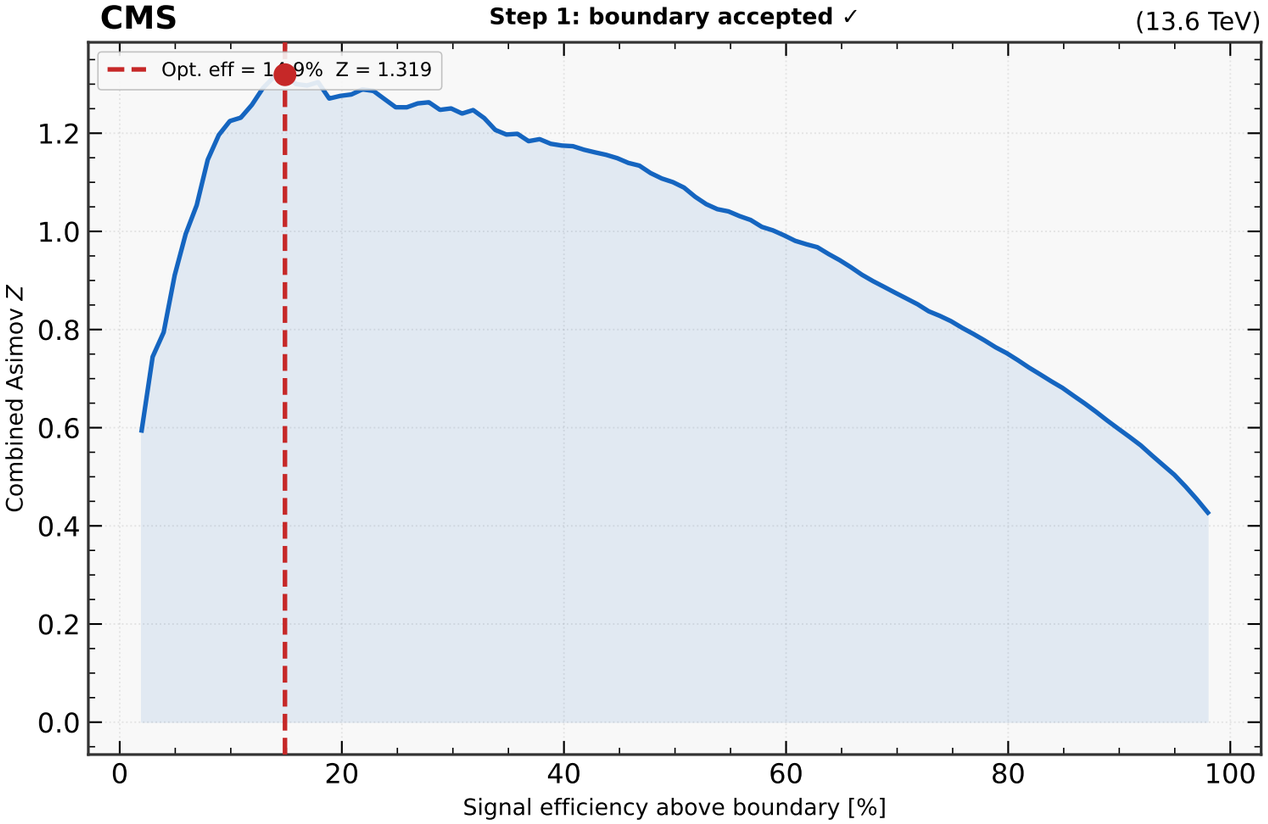
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.348	13392.57 ± 38.72	7629	0.00
[0.025, 0.050)	0.584	6423.87 ± 27.32	6300	0.01
[0.050, 0.075)	0.651	4334.29 ± 22.28	4551	0.02
[0.075, 0.100)	0.671	3240.45 ± 19.20	3533	0.02
[0.100, 0.125)	0.685	2577.73 ± 17.17	2899	0.03
[0.125, 0.150)	0.684	2132.46 ± 15.64	2322	0.03
[0.150, 0.175)	0.696	1781.77 ± 14.25	2048	0.04
[0.175, 0.200)	0.690	1517.77 ± 13.09	1760	0.05
[0.200, 0.225)	0.715	1325.15 ± 12.25	1516	0.05
[0.225, 0.250)	0.729	1213.43 ± 11.90	1399	0.06
[0.250, 0.275)	0.730	1077.54 ± 11.19	1200	0.07
[0.275, 0.300)	0.748	974.34 ± 10.57	1127	0.08
[0.300, 0.325)	0.761	895.04 ± 10.12	1043	0.08
[0.325, 0.350)	0.784	820.18 ± 9.70	945	0.10
[0.350, 0.375)	0.821	777.18 ± 9.53	937	0.11
[0.375, 0.400)	0.833	719.50 ± 9.15	833	0.12
[0.400, 0.425)	0.862	688.61 ± 8.95	776	0.12
[0.425, 0.450)	0.913	643.93 ± 8.72	774	0.14
[0.450, 0.475)	0.954	616.45 ± 8.53	730	0.15
[0.475, 0.500)	0.988	562.53 ± 8.09	734	0.18

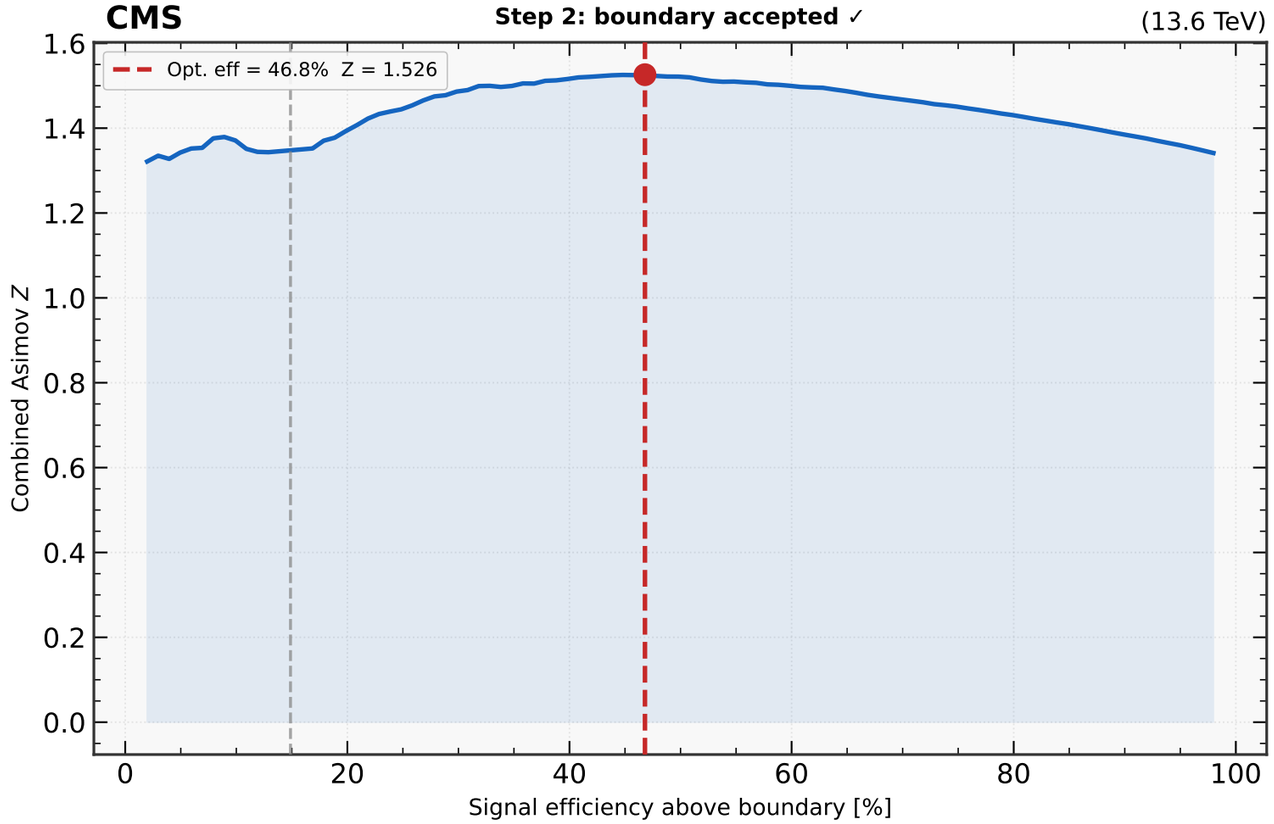
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

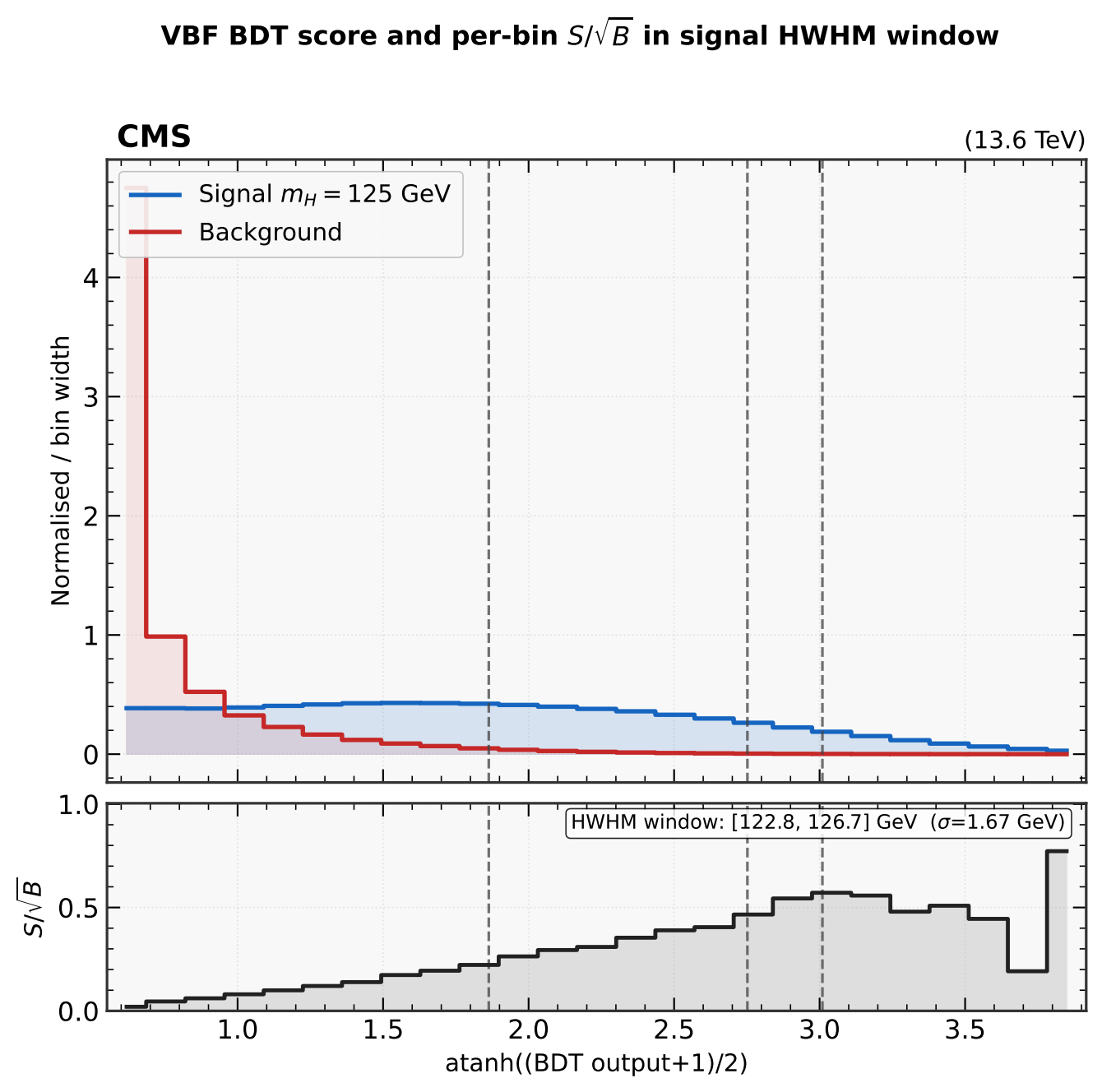
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	1.038	536.39 ± 7.88	601	0.19
[0.525, 0.550)	1.085	517.19 ± 7.77	613	0.21
[0.550, 0.575)	1.154	502.42 ± 7.63	602	0.23
[0.575, 0.600)	1.232	474.20 ± 7.32	581	0.26
[0.600, 0.625)	1.315	466.93 ± 7.36	626	0.28
[0.625, 0.650)	1.409	446.79 ± 7.15	526	0.31
[0.650, 0.675)	1.500	433.83 ± 7.06	556	0.34
[0.675, 0.700)	1.634	419.29 ± 6.94	498	0.39
[0.700, 0.725)	1.776	414.17 ± 6.83	524	0.43
[0.725, 0.750)	1.967	394.19 ± 6.73	552	0.50
[0.750, 0.775)	2.153	382.54 ± 6.55	519	0.56
[0.775, 0.800)	2.447	365.56 ± 6.42	505	0.66
[0.800, 0.825)	2.749	361.78 ± 6.37	536	0.75
[0.825, 0.850)	3.161	355.11 ± 6.32	487	0.88
[0.850, 0.875)	3.702	343.05 ± 6.17	493	1.07
[0.875, 0.900)	4.468	342.39 ± 6.26	457	1.29
[0.900, 0.925)	5.613	308.00 ± 5.81	495	1.79
[0.925, 0.950)	7.544	297.78 ± 5.74	455	2.47
[0.950, 0.975)	11.466	266.27 ± 5.38	452	4.13
[0.975, 1.000)	19.573	146.42 ± 3.80	306	11.79

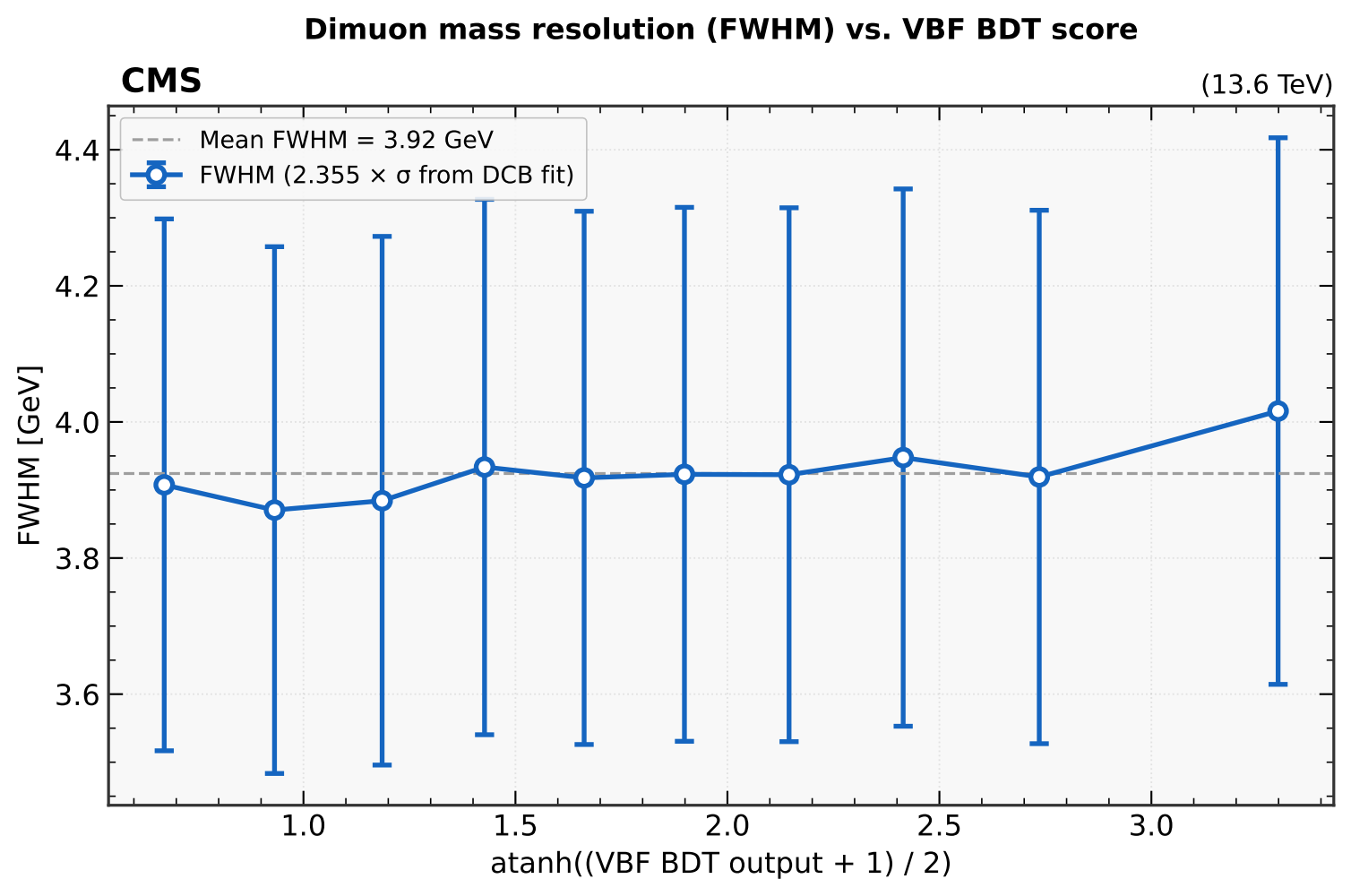

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.5781
0.316
1.319
1.526
1.578
Combined Asimov Z [HWHM window]
Number of BDT categories
1.6
1.4
1.2
1.0
0.8
0.6
0.4
1
2
3
4

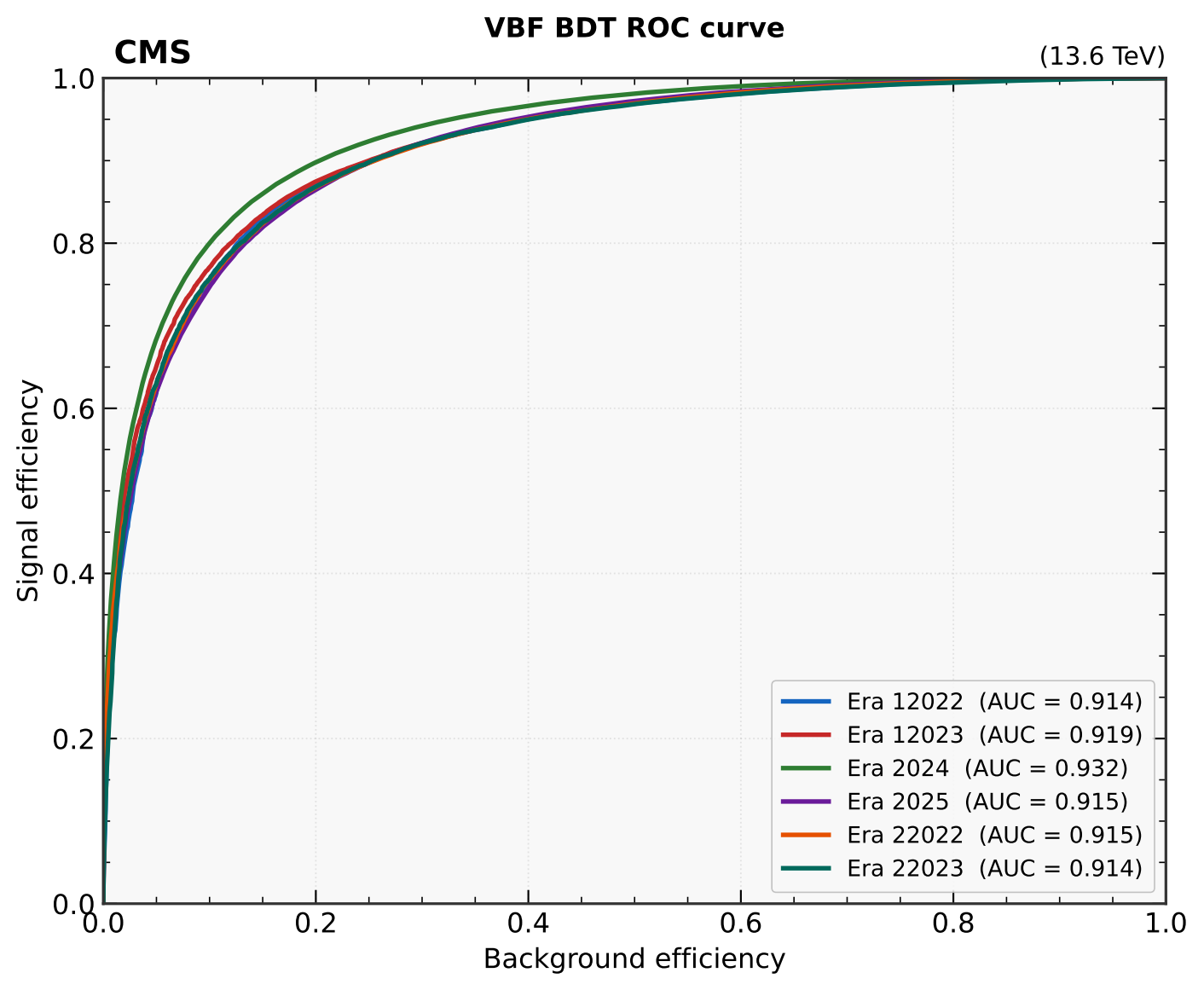
Iterative category boundary optimisation (AN Fig. 32 analog)

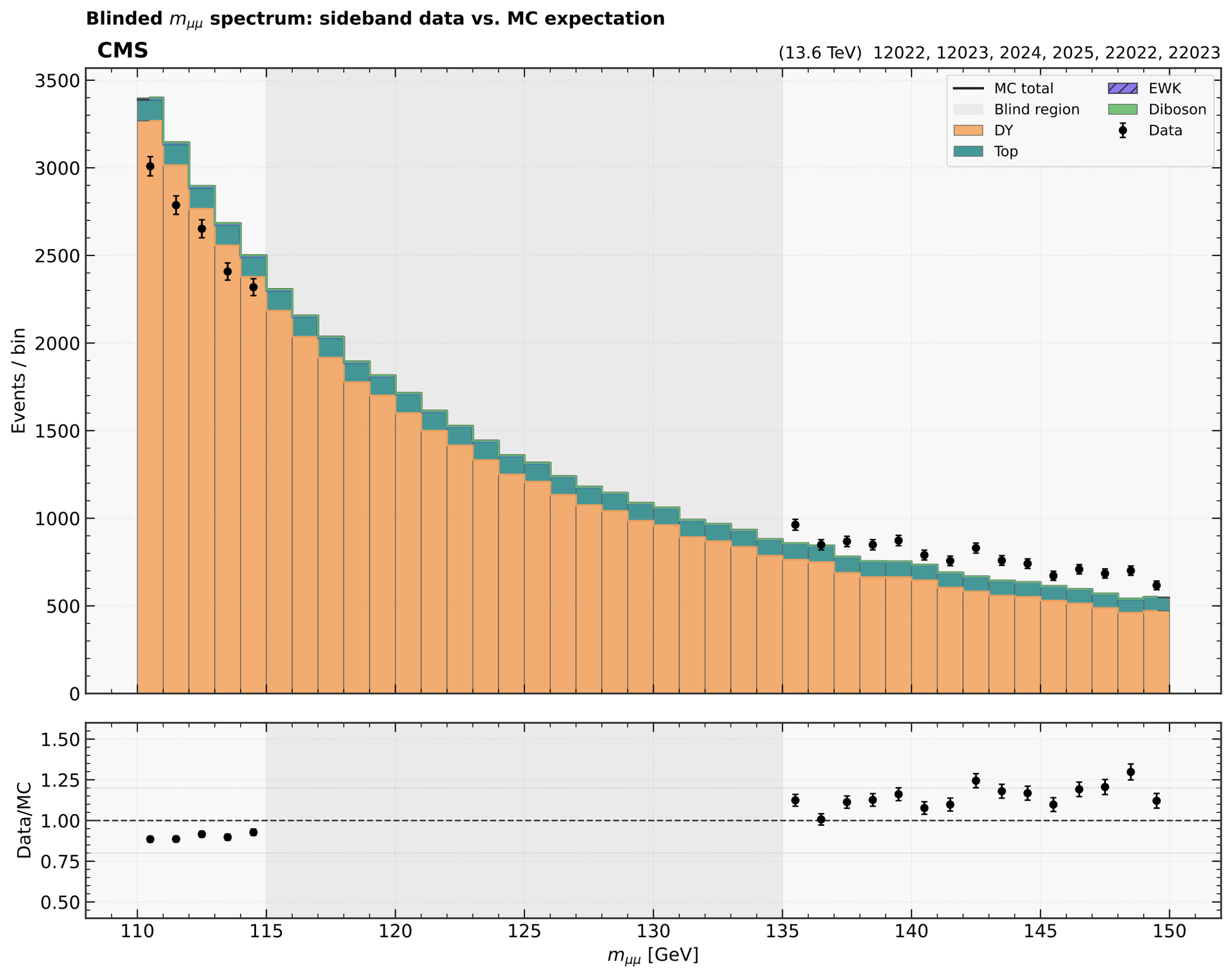


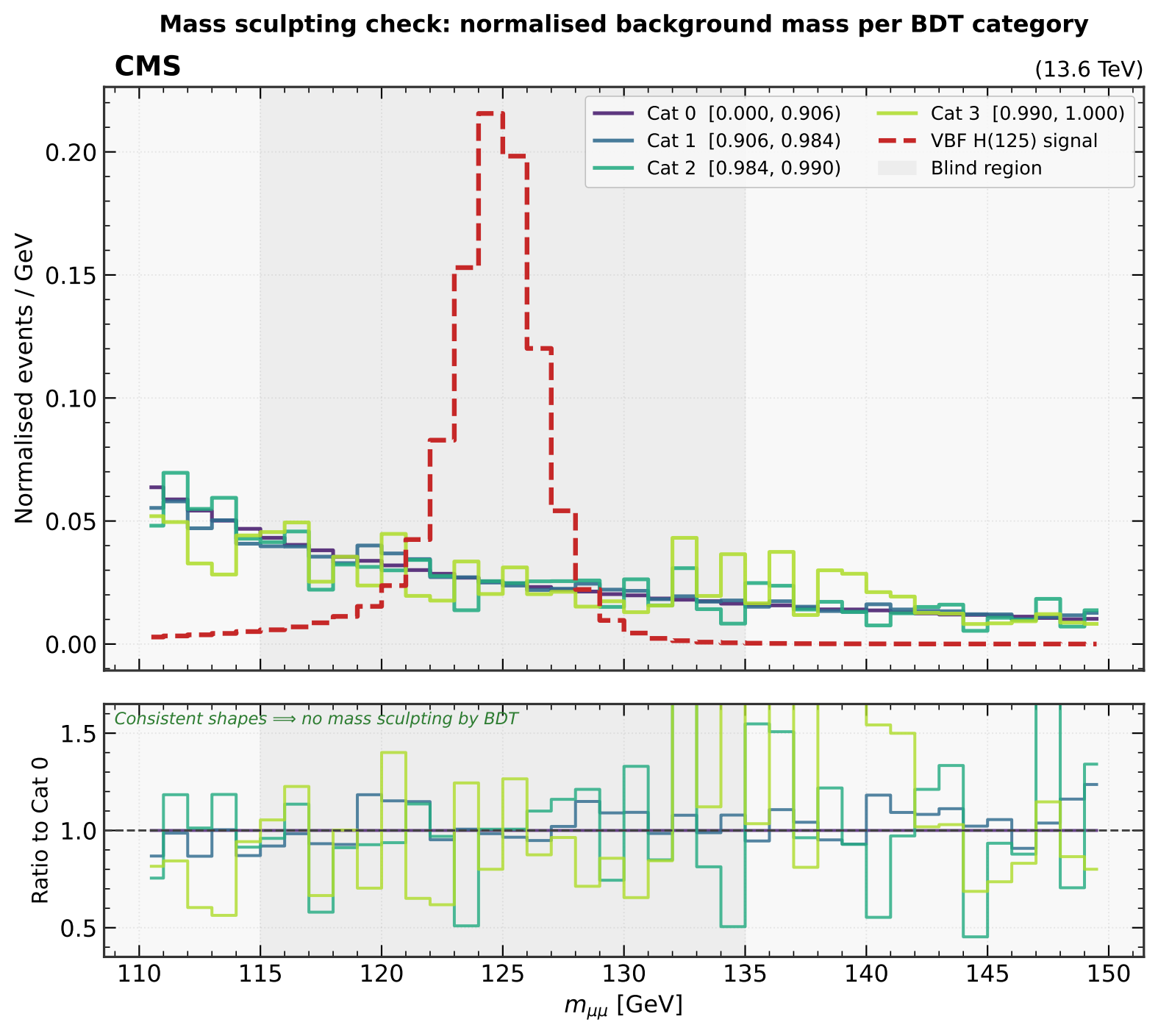



VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window
CMS
(13.6 TeV)
Signal $m_H = 125$ GeV
Background
Normalised / bin width
0
1
2
3
4
HWHM window: [122.8, 126.7] GeV (σ=1.67 GeV)
$S/\sqrt{B}$
0.0
0.5
1.0
1.0
1.5
2.0
2.5
3.0
3.5
atanh((BDT output+1)/2)

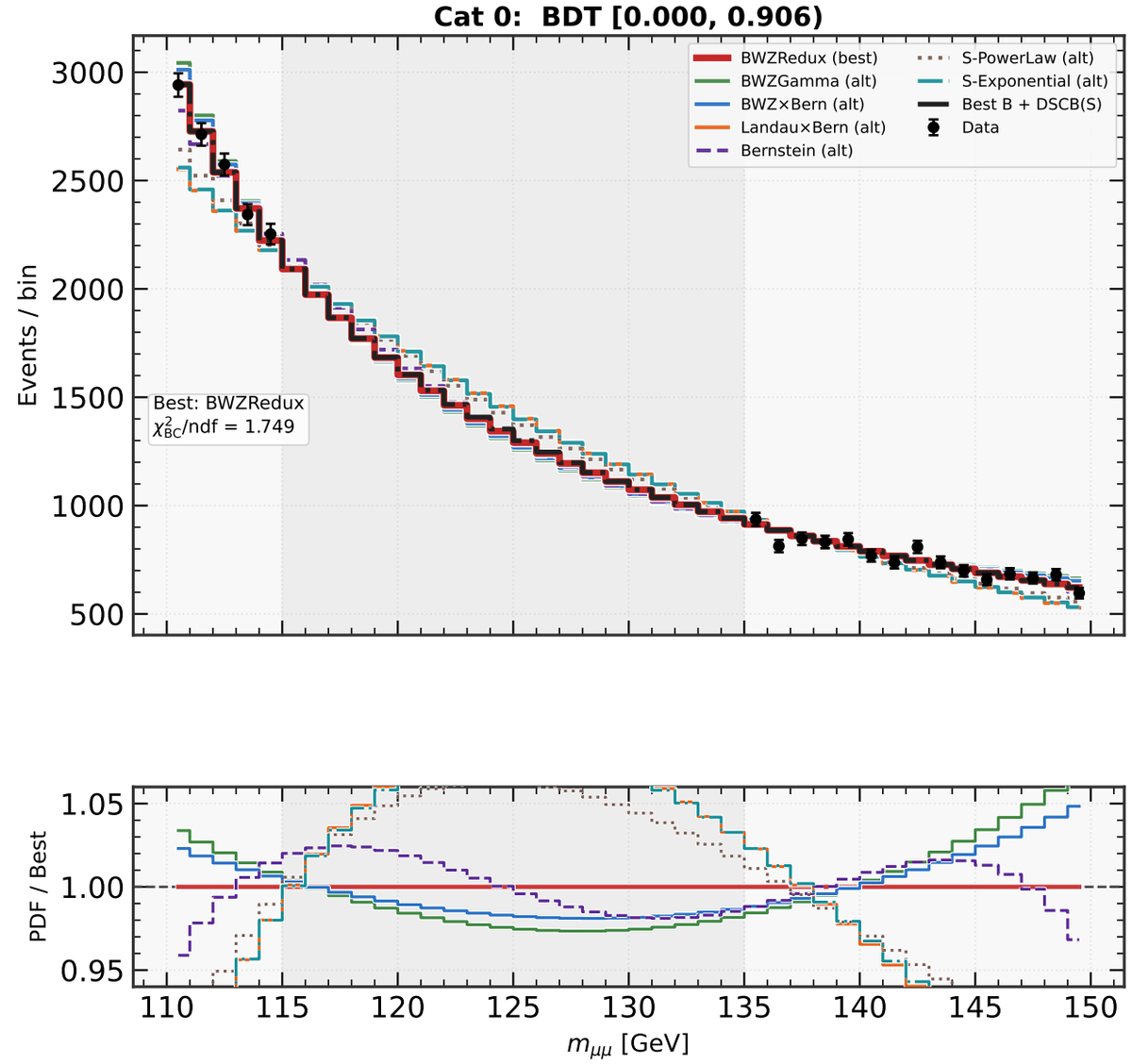

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.92 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

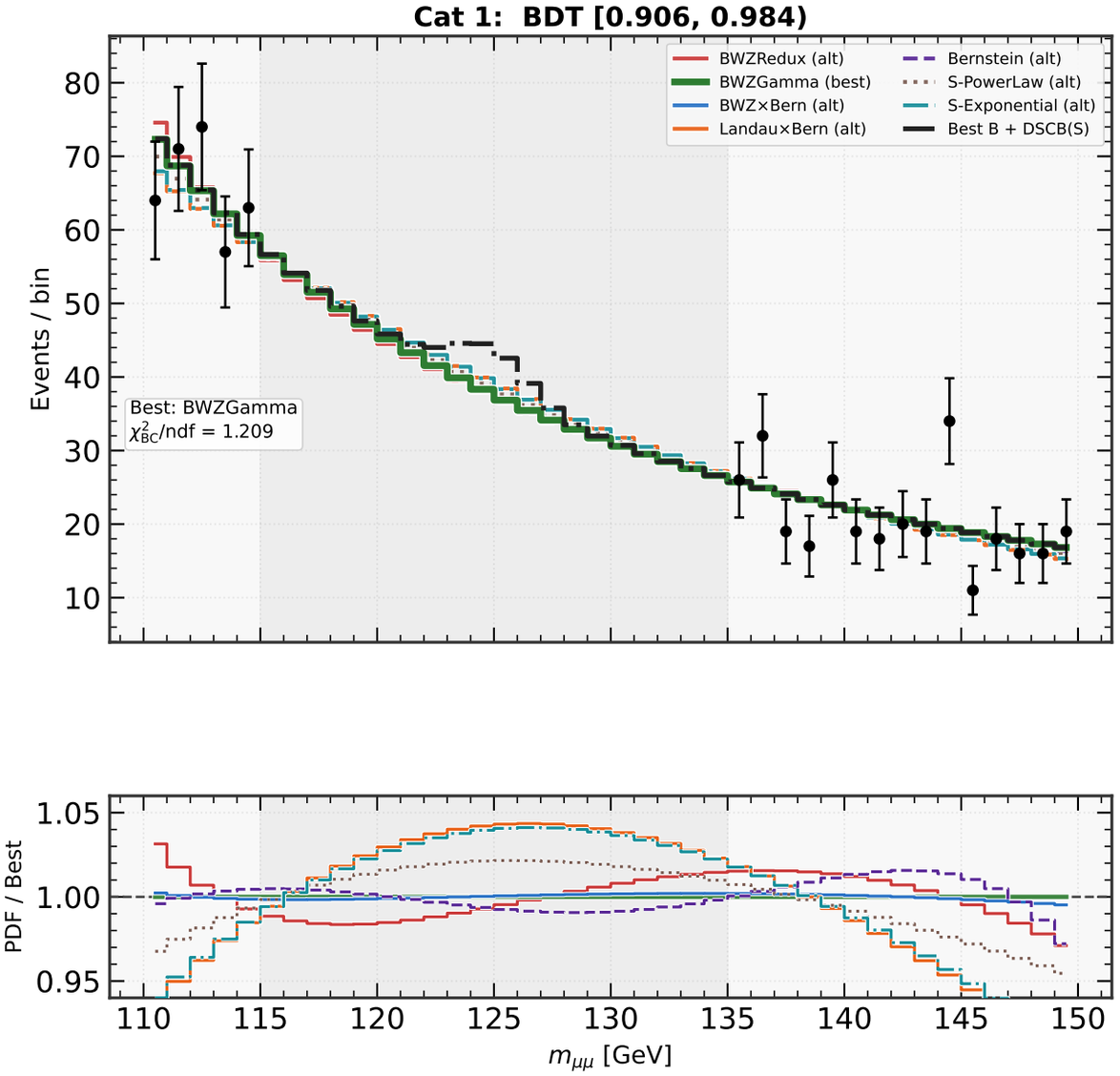

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.914)
Era 12023 (AUC = 0.919)
Era 2024 (AUC = 0.932)
Era 2025 (AUC = 0.915)
Era 22022 (AUC = 0.915)
Era 22023 (AUC = 0.914)

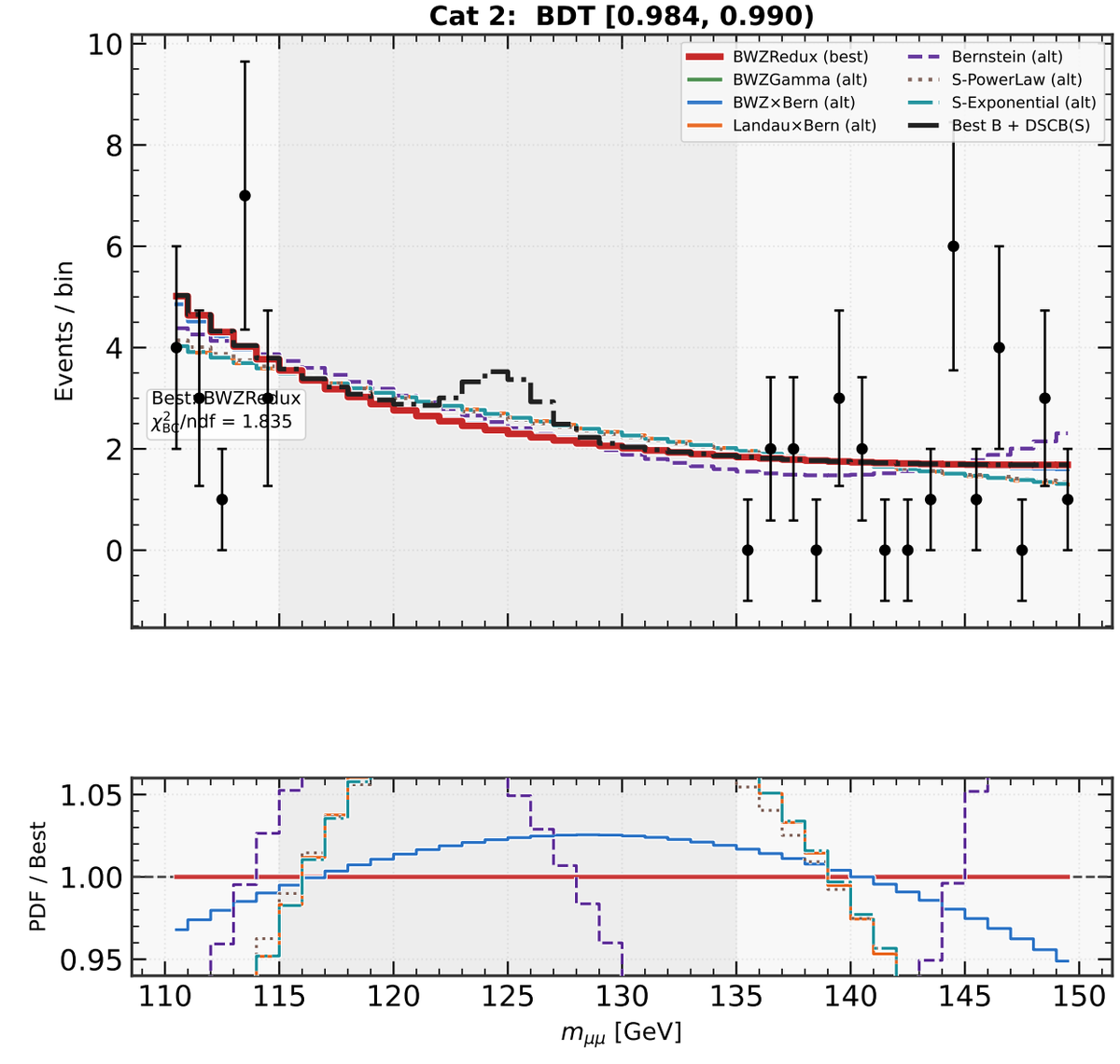

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022, 12023, 2024, 2025, 22022, 22023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

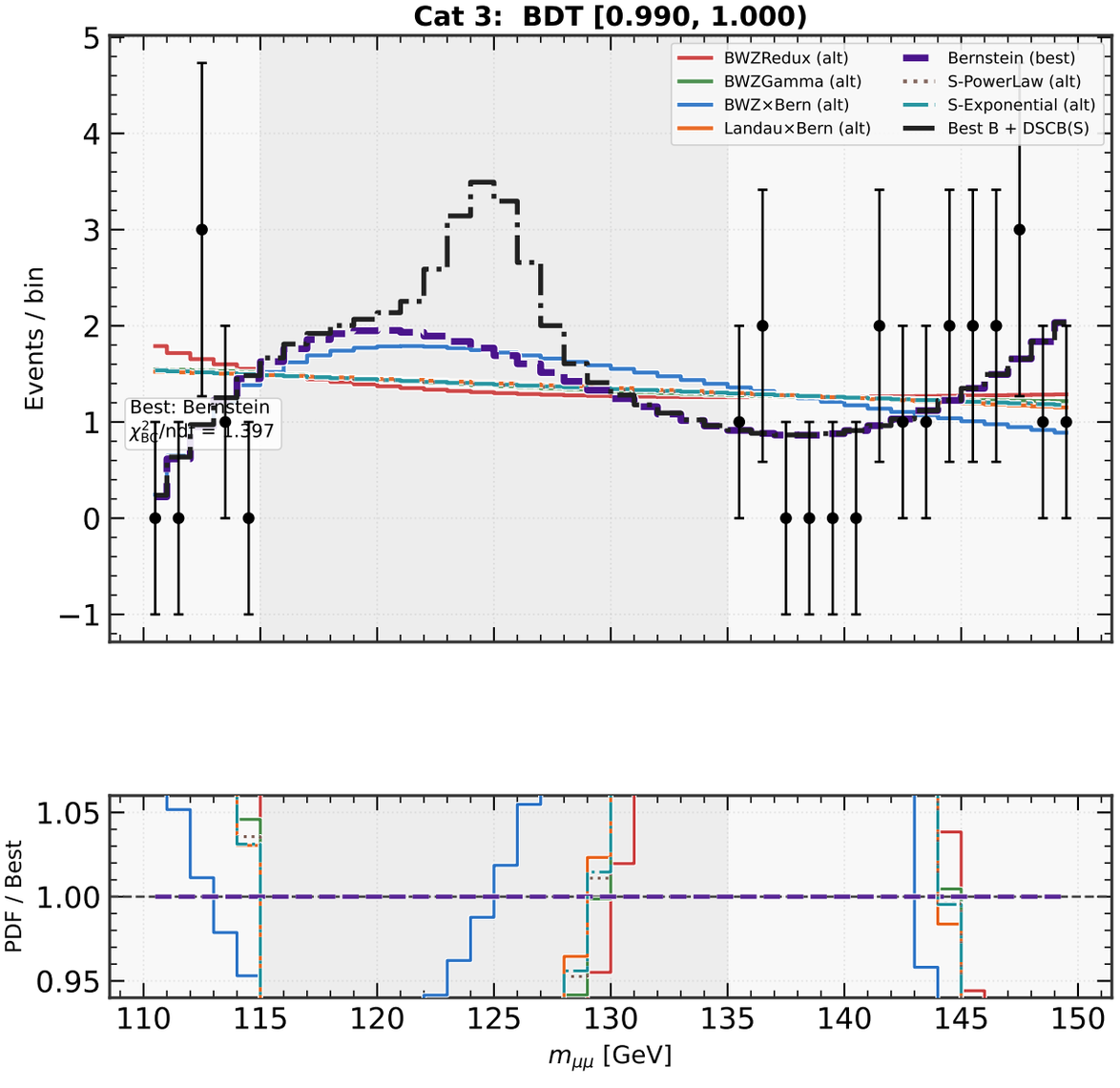

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.906)
Cat 1 [0.906, 0.984)
Cat 2 [0.984, 0.990)
Cat 3 [0.990, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits





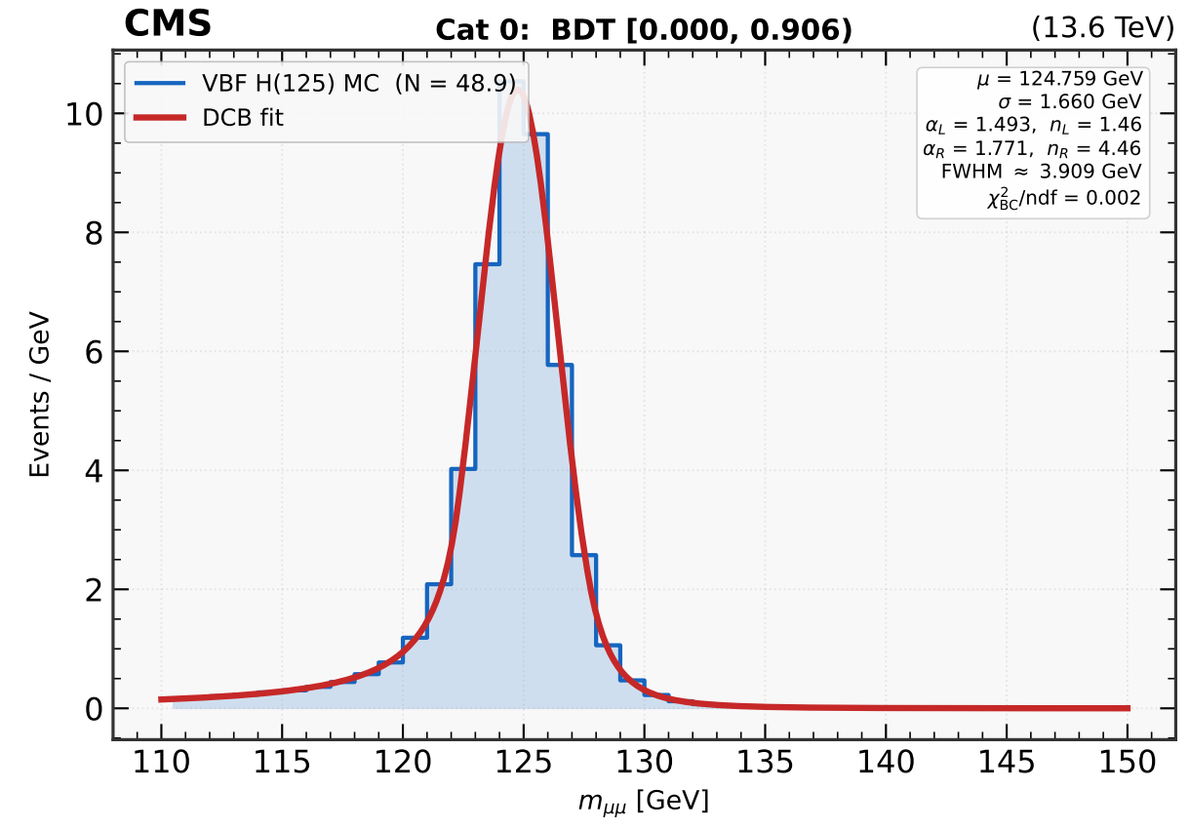


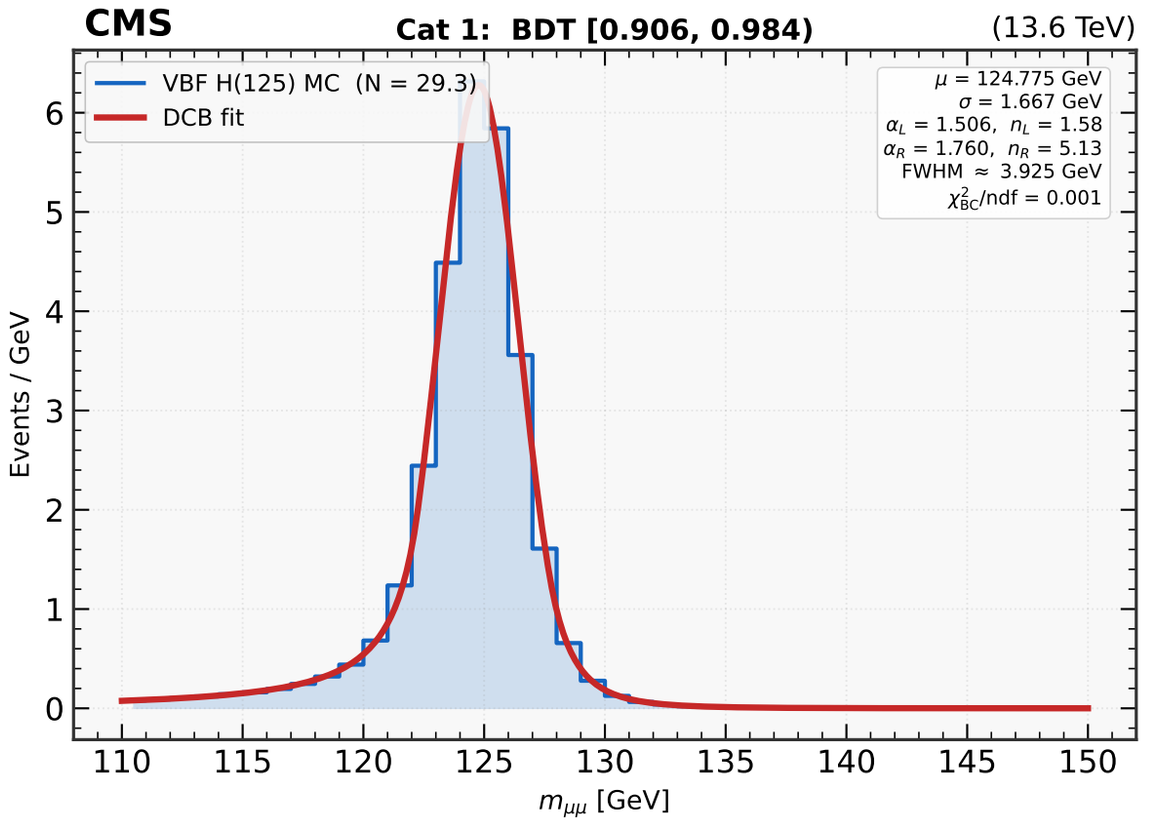


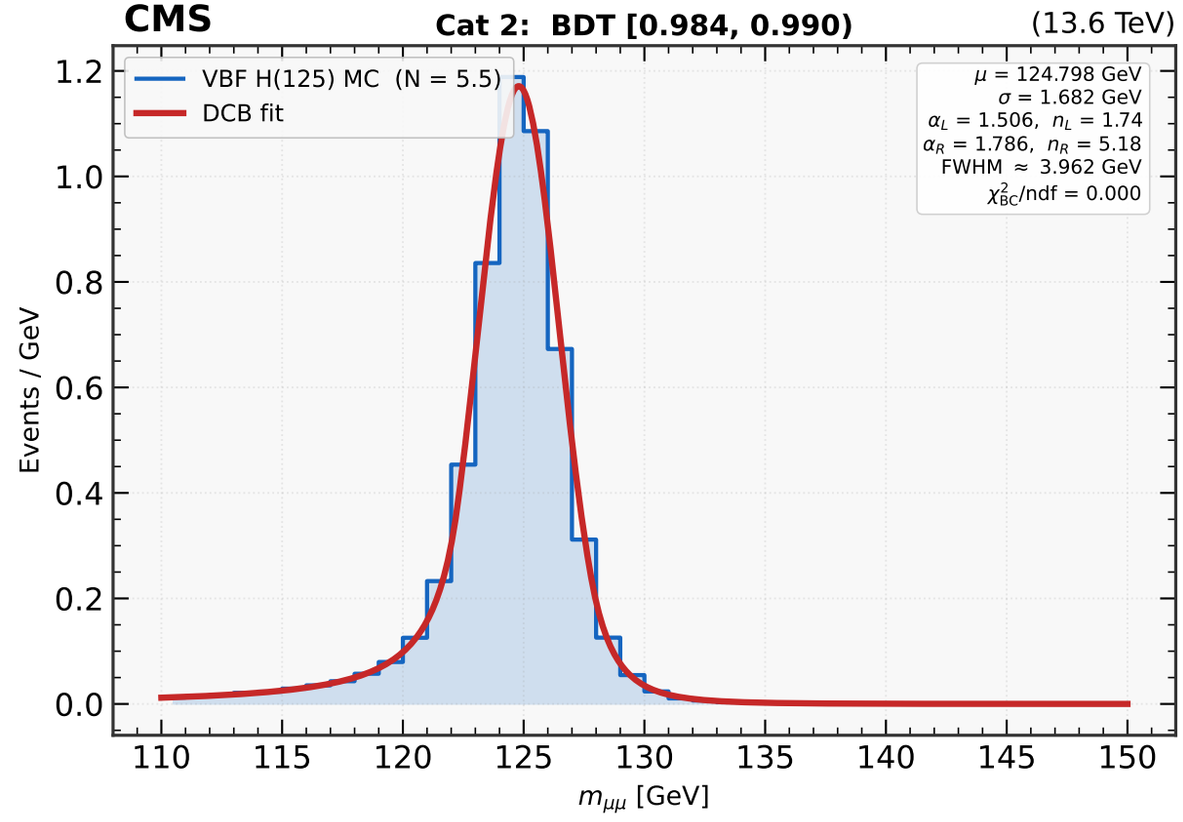
Per-category background model fit comparison (χ²_BC/ndf, ndf)

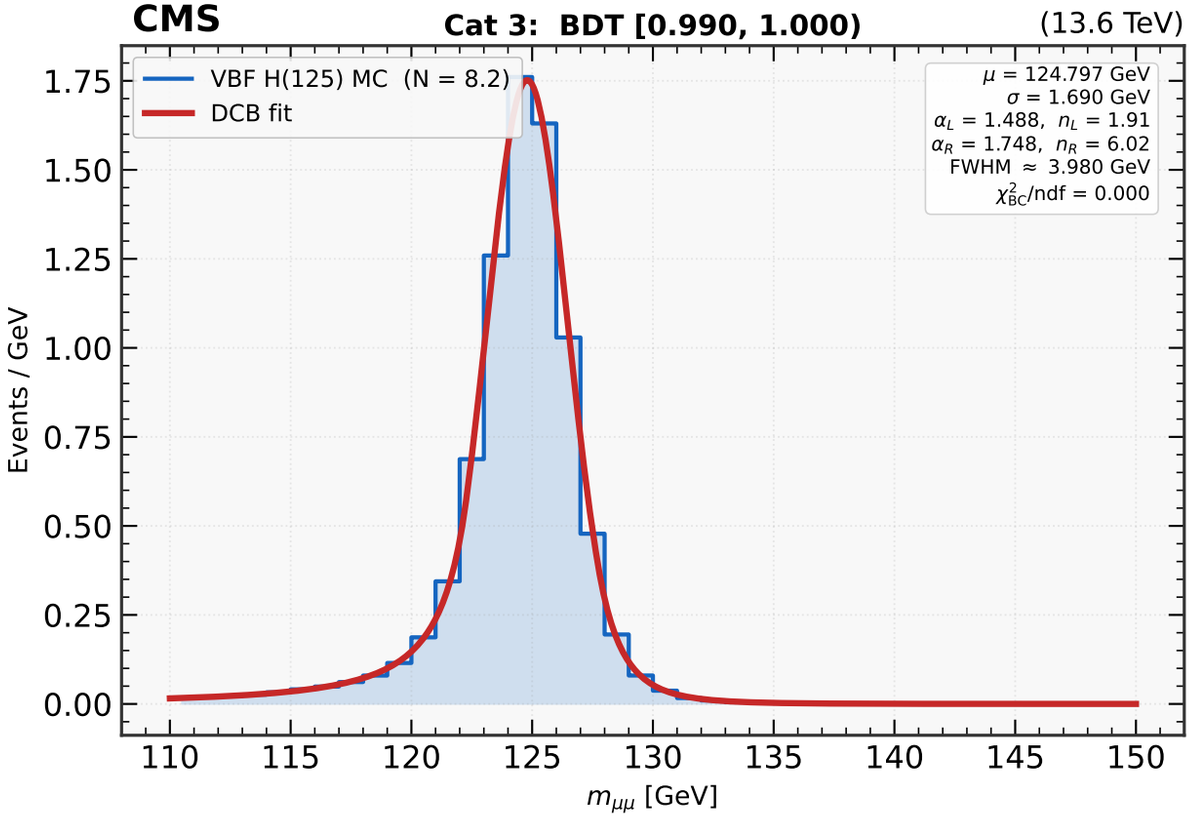
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.906)	BWZRedux	1.749	1.749 (37)	2.480 (37)	2.170 (36)	9.303 (37)	2.332 (35)	5.812 (37)	8.906 (37)
1	[0.906, 0.984)	BWZGamma	1.209	1.219 (37)	1.209 (37)	1.244 (36)	1.270 (37)	1.278 (35)	1.226 (37)	1.264 (37)
2	[0.984, 0.990)	BWZRedux	1.835	1.835 (37)	1.837 (37)	1.888 (36)	1.852 (37)	1.863 (35)	1.844 (37)	1.851 (37)
3	[0.990, 1.000)	Bernstein	1.397	1.533 (37)	1.511 (37)	1.501 (36)	1.519 (37)	1.397 (34)	1.518 (37)	1.518 (37)

VBF H(125) signal DCB extraction per BDT category



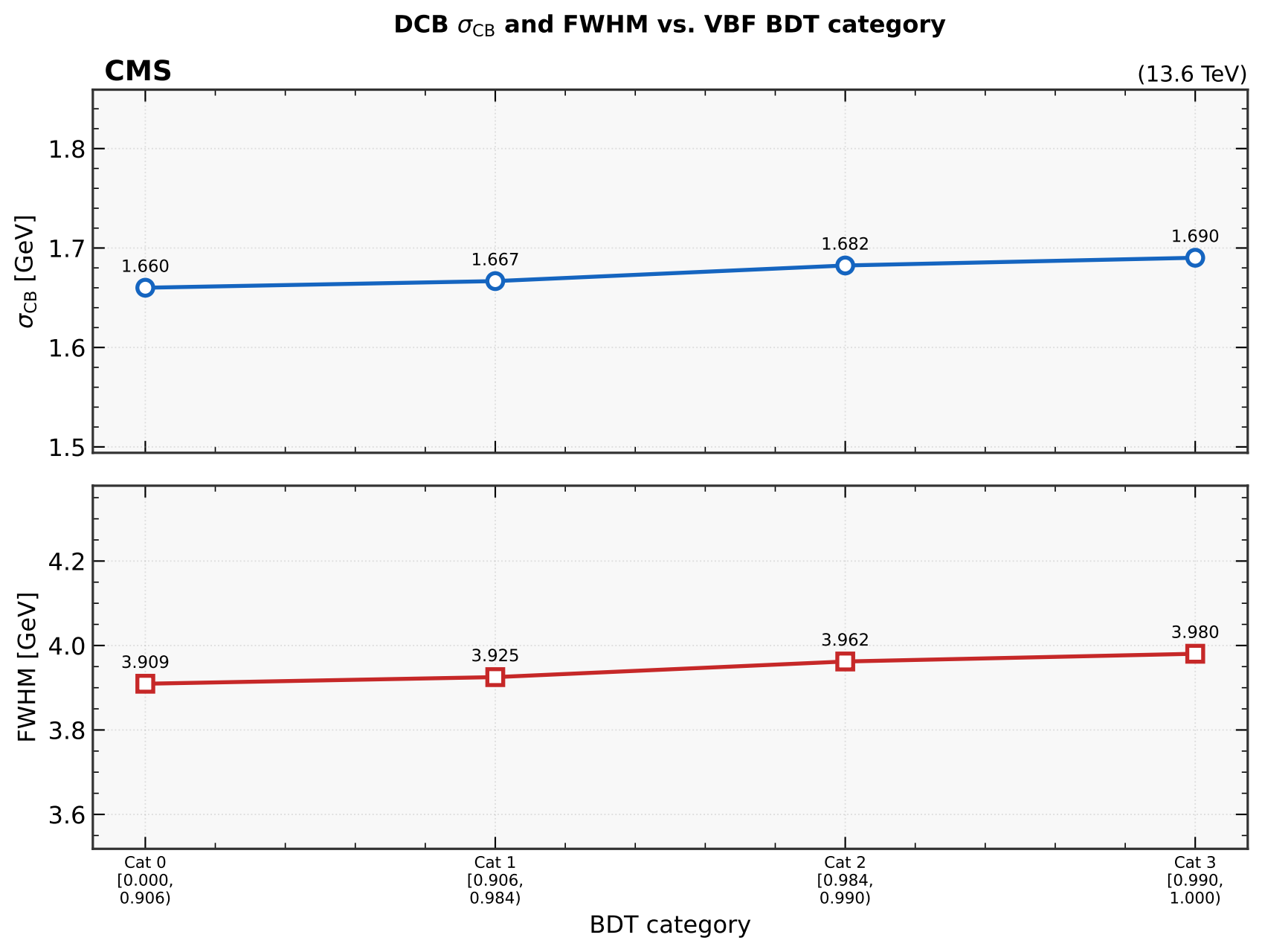


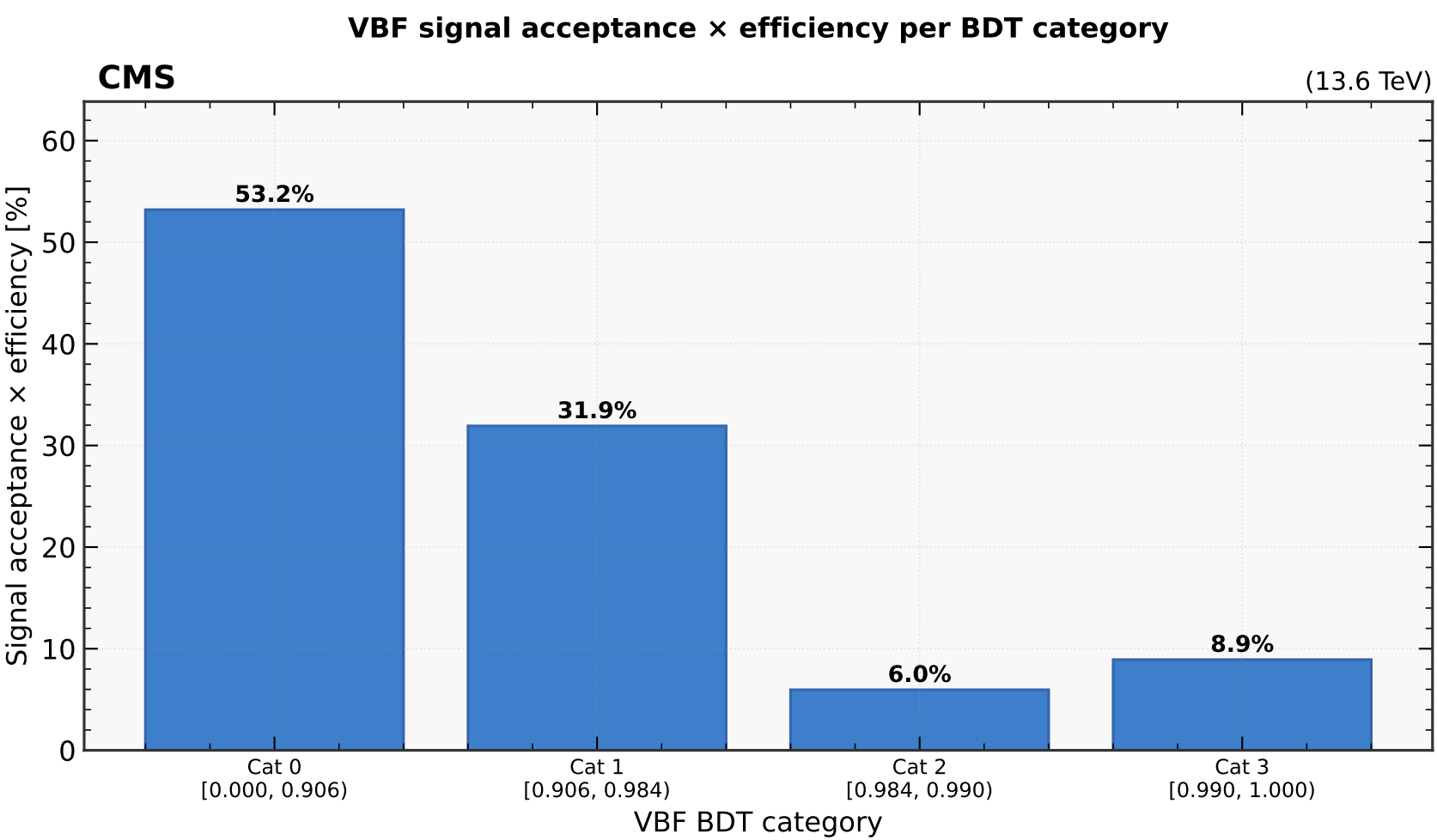




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.906)	48.86	124.7587	1.6601	1.4932	1.459	1.7712	4.455	3.909	0.169	0.002
1	[0.906, 0.984)	29.31	124.7748	1.6667	1.5059	1.579	1.7598	5.126	3.925	0.796	0.001
2	[0.984, 0.990)	5.47	124.7978	1.6824	1.5059	1.740	1.7858	5.176	3.962	0.696	0.000
3	[0.990, 1.000)	8.19	124.7967	1.6902	1.4878	1.906	1.7478	6.020	3.980	1.167	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.5
1.660
1.667
1.682
1.690
FWHM [GeV]
4.2
4.0
3.8
3.6
3.909
3.925
3.962
3.980
Cat 0 [0.000, 0.906)
Cat 1 [0.906, 0.984)
Cat 2 [0.984, 0.990)
Cat 3 [0.990, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
53.2%
31.9%
6.0%
8.9%
Cat 0
[0.000, 0.906)
Cat 1
[0.906, 0.984)
Cat 2
[0.984, 0.990)
Cat 3
[0.990, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.906)	48.86	389225.54	33.183	38615.713	[122.8, 126.7]	0.169	0.169
1	[0.906, 0.984)	29.31	6476.54	20.094	630.410	[122.8, 126.7]	0.800	0.796
2	[0.984, 0.990)	5.47	317.59	3.778	28.232	[122.8, 126.8]	0.711	0.696
3	[0.990, 1.000)	8.19	214.05	5.687	21.916	[122.8, 126.8]	1.215	1.167

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.906)	24134	180813.23 ± 390.45	0.133	389225.54 ± 573.67	0.1689	0.1688	38615.713
1	[0.906, 0.984)	639	2930.15 ± 48.43	0.218	6476.54 ± 72.50	0.8003	0.7961	630.410
2	[0.984, 0.990)	43	153.66 ± 10.73	0.280	317.59 ± 15.19	0.7109	0.6959	28.232
3	[0.990, 1.000)	22	96.63 ± 7.27	0.228	214.05 ± 11.06	1.2147	1.1671	21.916
Combined	all						1.5839	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.906)	359895.01	635.42	27975.35	719.75	329101.21	389225.54
1	[0.906, 0.984)	5740.99	304.10	410.20	21.24	5256.56	6476.54
2	[0.984, 0.990)	249.14	50.56	16.97	0.93	230.77	317.59
3	[0.990, 1.000)	144.68	58.65	10.31	0.41	122.35	214.05